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Thermal Simulation of a Showerhead Injector Operating with Hydrogen Peroxide

3D Transient Heat-Up Analysis of an HTP-Cooled Injector Face Plate

Modeling of Combustion Processes (MKWS)

This report describes the development of a three-dimensional transient thermal simulation of a *showerhead*-type rocket injector operating with high-test hydrogen peroxide (HTP) as the oxidiser in a HTP / ethanol bipropellant configuration. The objective is to predict the temperature distribution within the injector face plate during combustion-chamber operation and to determine the temperature at which a (quasi-)steady state is established. A unit-cell (sector) model around a single injector orifice is solved with a fully three-dimensional explicit finite-volume conduction scheme implemented in NumPy. The hot combustion gases heat the chamber-side face by convection and radiation, while the liquid HTP flowing through the orifices removes heat in a quasi-regenerative manner. The simulation predicts a strongly non-uniform face temperature, from about $393\,°C$ at the cooled channel wall to about $849\,°C$ in the metal between orifices, with a heat-up time constant of approximately 1.0 s.

Contents

1 Introduction

Injector face plates in liquid rocket engines are exposed to extreme thermal loads: the chamber side is heated by hot combustion gases, while the propellant manifold and the internal flow passages are bathed in relatively cold liquid propellant. The balance between gas-side heating and propellant-side cooling sets the operating temperature of the plate, which is critical both for its structural integrity and for the integrity of the propellant itself [1].

This project develops an internal thermal simulator for a *showerhead* injector running on **hydrogen peroxide (HTP)** used as the oxidiser, with ethanol as the fuel. In contrast to a one-dimensional through-thickness estimate, a fully three-dimensional model is built so that the heat flow around the individual cooling orifices – and therefore the in-plane temperature non-uniformity of the face – is resolved.

1.1 Project Objectives

- Build a 3D transient conduction model of a showerhead injector unit cell.
- Represent the combined gas-side convective/radiative heating and the quasi-regenerative cooling by the HTP flowing through the orifices.
- Determine the steady-state temperature field and the heat-up time constant of the plate.
- Provide a parametric, dependency-light implementation (NumPy only) with reproducible figures and an animation of the heat-up.

2 Injector and Operating Case

2.1 Showerhead Injector Concept

A showerhead injector introduces the propellants through a regular pattern of straight axial orifices drilled in a face plate. The plate considered here has $n_{\text{holes}} = 60$ orifices distributed over a face of radius R. Because the thermal boundary conditions are essentially uniform over the face, the plate is geometrically periodic and can be represented by a single repeating **unit cell** (sector) surrounding one orifice. The in-plane footprint per orifice equals $A_{\text{face}}/n_{\text{holes}}$, which defines the cell pitch

$$p = \sqrt{\frac{A_{\text{face}}}{n_{\text{holes}}}} = \sqrt{\frac{\pi R^2}{n_{\text{holes}}}}. \tag{1}$$

2.2 Propellant Combination

The engine uses high-test hydrogen peroxide as the oxidiser and ethanol as the fuel – a *green*, storable combination that avoids the toxic hydrazine / nitrogen-tetroxide propellants of earlier manoeuvring engines [3]. For the present heat-transfer study the relevant data are the liquid HTP properties (used to size the channel cooling) and the combustion-gas recovery temperature (used as the hot-side driving temperature). The HTP / ethanol flame temperature is taken as $T_{\text{gas}} = 2500\,\text{K}$.

2.3 Thermal Loading Scenario

During steady chamber operation the plate experiences:

- gas-side heating of the chamber face ($z = 0$) by convection and radiation;
- internal cooling of each orifice wall by the through-flowing liquid HTP;
- cooling of the manifold face ($z = L$) by the cold inlet HTP.

The lateral faces of the unit cell are adiabatic, representing the mirror symmetry between neighbouring orifices.

3 Computational Model

3.1 Heat Conduction Equation

Within the solid metal the temperature $T(\mathbf{x}, t)$ obeys the transient heat-conduction equation

$$\rho\, c_p\, \frac{\partial T}{\partial t} = k\, \nabla^2 T = k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right), \tag{2}$$

with density ρ, specific heat c_p and conductivity k; the thermal diffusivity is $\alpha = k/(\rho c_p)$.

3.2 Gas-Side Boundary Condition

The chamber-side face receives a combined convective and radiative flux

$$q''_{\text{gas}} = h_{\text{gas}}\,(T_{\text{gas}} - T) + \varepsilon\,\sigma\,(T_{\text{gas}}^4 - T^4), \tag{3}$$

with $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}K^{-4}}$. The film coefficient h_{gas} is the dominant uncertainty and is treated as an input parameter.

3.3 Channel Cooling – Dittus–Boelter

The convective coefficient inside an orifice is obtained from the Dittus–Boelter correlation for a fluid being heated in turbulent pipe flow [2]:

$$\mathrm{Nu} = 0.023\,\mathrm{Re}^{0.8}\,\mathrm{Pr}^{0.4}, \qquad h_p = \frac{\mathrm{Nu}\,k_{\text{prop}}}{d_{\text{hole}}}, \tag{4}$$

with $\mathrm{Re} = \rho_{\text{prop}}\, v\, d_{\text{hole}}/\mu$ and $\mathrm{Pr} = c_{p,\text{prop}}\,\mu/k_{\text{prop}}$. The orifice wall flux is $q''_{\text{ch}} = h_p\,(T - T_{\text{prop}})$ and the manifold face is cooled by $q''_{\text{back}} = h_{\text{back}}\,(T - T_{\text{prop}})$.

3.4 Numerical Scheme and Stability

Equation (2) is discretised with a cell-centred finite-volume scheme on a uniform Cartesian grid $N_x \times N_y \times N_z$; the cylindrical orifice is a "void" region cut from the mesh (staircase approximation) whose faces carry the convective flux instead of conduction. Time integration uses the explicit forward-time, centred-space (FTCS) scheme, which is conditionally stable with the step

$$\Delta t = \frac{\mathrm{SF}}{2\alpha\left(\dfrac{1}{\Delta x^2} + \dfrac{1}{\Delta y^2} + \dfrac{1}{\Delta z^2}\right) + \dfrac{(h_{\text{gas}} + h_{\text{back}})A_z + 2h_p(A_x + A_y) + 4\varepsilon\sigma T_{\text{gas}}^3 A_z}{\rho c_p V}}, \tag{5}$$

with a safety factor $\mathrm{SF} = 0.5$. A (quasi-)steady state is detected when $\max|\Delta T/\Delta t| < 5 \times 10^{-3}\,\mathrm{K/s}$, and the heat-up time constant τ is the time to reach 63.2% of the total temperature rise of the hottest probe.

3.5 Material and Propellant Properties

The baseline plate is 316L stainless steel; the cooling HTP is taken at ~90 % concentration. The properties are listed in Table 1.

Table 1: Solid (316L) and liquid HTP properties used in the simulation.

Quantity	Symbol	Value
Plate conductivity	k	16.0 $\mathrm{W\,m^{-1}K^{-1}}$
Plate density	ρ	8000 $\mathrm{kg\,m^{-3}}$
Plate specific heat	c_p	500 $\mathrm{J\,kg^{-1}K^{-1}}$
Plate diffusivity	α	4.0×10^{-6} $\mathrm{m^2\,s^{-1}}$
HTP density	ρ_{prop}	1390 $\mathrm{kg\,m^{-3}}$
HTP specific heat	$c_{p,\mathrm{prop}}$	2730 $\mathrm{J\,kg^{-1}K^{-1}}$
HTP conductivity	k_{prop}	0.50 $\mathrm{W\,m^{-1}K^{-1}}$
HTP viscosity	μ	1.1×10^{-3} Pa s
HTP Prandtl number	Pr	6.01 –
HTP inlet temperature	T_{in}	20 °C

4 Implementation

4.1 Solver Architecture

The simulator is a single Python 3 program depending only on NumPy and Matplotlib (no SciPy). The temperature field is stored as a 3D array; the solid metal and the orifice void are distinguished by a boolean mask. All face fluxes (conduction, channel convection, gas-side and manifold boundary terms) are evaluated with vectorised array slicing, so one explicit time step is a handful of NumPy operations.

4.2 Unit-Cell Update

Listing 1 shows the core of the explicit update for the conduction and channel-wall convection in the x-direction; the y- and z-directions are analogous.

```
# conduction between adjacent solid cells (x-direction)
both = solid[:-1] & solid[1:]
fc = np.where(both, Gx * (T[1:] - T[:-1]), 0.0)
Q[:-1] += fc; Q[1:] -= fc
# channel wall: solid cell facing the void (HTP) -> convective cooling
lsv = solid[:-1] & ~solid[1:]
Q[:-1] += np.where(lsv, Hx * (T_prop - T[:-1]), 0.0)
rsv = ~solid[:-1] & solid[1:]
Q[1:]  += np.where(rsv, Hx * (T_prop - T[1:]), 0.0)
# explicit time advance of the solid cells
Tn = T + (dt / Cap) * Q
Tn[~solid] = T_prop
```

Listing 1: Vectorised flux assembly (excerpt).

4.3 Simulation Configuration

The gas-side, operating and geometric parameters of the baseline run are given in Table 2.

4.4 Reproducibility

The full simulation, post-processing and visualisation run with a single command, `python3 injector_thermal_sim.py`, producing the figures used in this report and an animated GIF of the heat-up. A 10 s transient completes in about 25 s on a laptop.

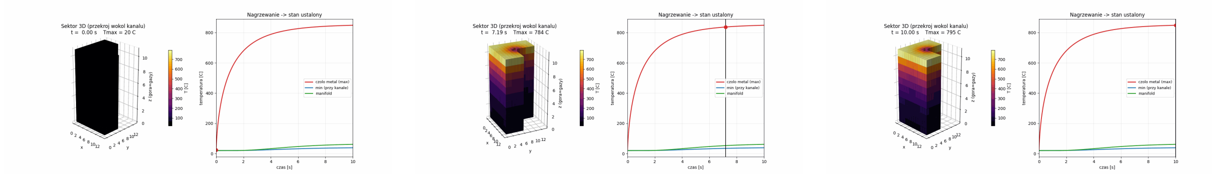
Table 2: Baseline simulation configuration.

Quantity	Symbol	Value
Flame / recovery temperature (HTP+ethanol)	T_{gas}	2500 K
Gas-side film coefficient	h_{gas}	2000 $\mathrm{W\,m^{-2}K^{-1}}$
Emissivity	ε	0.30 –
Manifold cooling coeff.	h_{back}	1500 $\mathrm{W\,m^{-2}K^{-1}}$
Total HTP mass flow	$\dot{m}$	0.5 $\mathrm{kg\,s^{-1}}$
Plate radius	R	30 mm
Plate thickness	L	12 mm
Number of orifices	n_{holes}	60 –
Orifice diameter	d_{hole}	1.5 mm
Cell pitch (Eq. 1)	p	6.86 mm
Grid	$N_x \times N_y \times N_z$	26 × 26 × 22
Time step	Δt	1.54 ms

5 Simulation Results

5.1 3D Temperature Field

Figure 1 shows the evolution of the unit-cell temperature field, displayed as a cut-away around the cooling channel (gas side on top). The plate heats from a uniform cold start to a state with a hot gas-side face and a cold manifold side. The full animation is provided as the supplementary file `injector_thermal_3d.gif`.



(a) $t = 0$ (cold start) (b) intermediate heat-up (c) near steady state

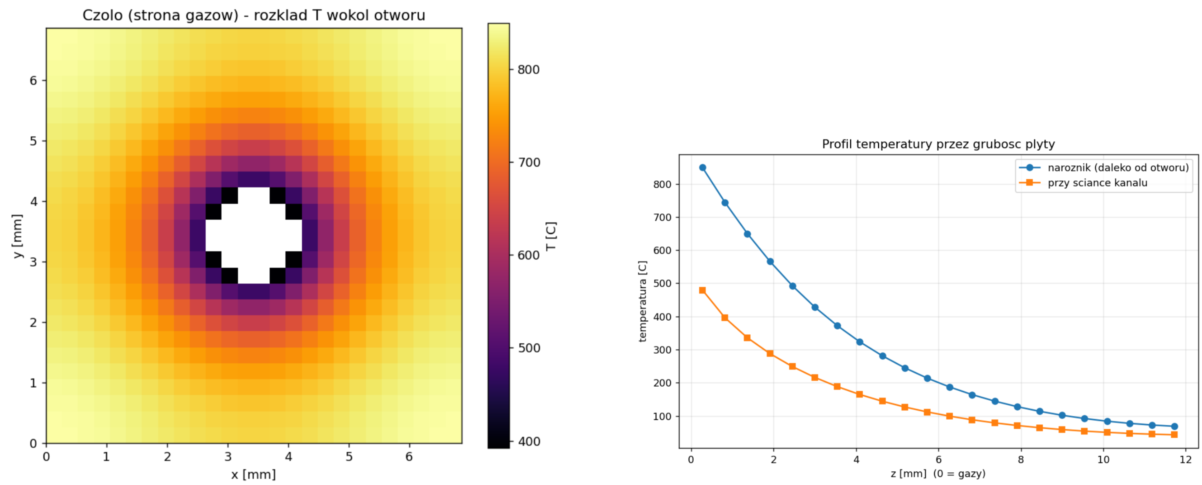
Figure 1: Three-dimensional unit-cell temperature field at three instants (cut-away around the orifice).

5.2 Gas-Side Face Distribution

The most important result is the strongly non-uniform face temperature (Fig. 2a): the metal adjacent to the cooled channel wall stabilises near $393\,°C$, whereas the metal midway between orifices reaches $849\,°C$, a difference of about 457 K across a single pitch. This in-plane gradient is exactly the three-dimensional effect that motivates resolving the orifice geometry.

5.3 Transient Behaviour and Time Constant

Figure 3 shows the temperature histories of the characteristic probes. The hot face responds quickly, with a time constant $\tau \approx 1.0\,\mathrm{s}$, and is essentially at its operating value within a few seconds. The manifold side and the channel-wall region drift more slowly because of the low conductivity of stainless steel, so the *whole-body* steady state is approached only towards the end of the 10 s window even though the thermally critical gas-side face has already levelled off.



(a) Gas-side face: distribution around the orifice.

(b) Through-thickness profiles.

Figure 2: Steady-state temperature map of the gas-side face and through-thickness profiles at the corner (far from the orifice) and next to the channel wall.

5.4 Output Parameters

The key output quantities are summarised in Table 3.

Table 3: Simulation output parameters (316L plate, baseline case).

Parameter	Value
Face metal, maximum (between orifices)	849.2 °C
Face metal, minimum (at channel wall)	392.6 °C
In-plane gradient across the face	456.6 K
Manifold side (mean)	61.2 °C
Channel heat-transfer coefficient h_p	$17\,484\ \mathrm{W\,m^{-2}K^{-1}}$
Heat removed per orifice	157.7 W
HTP bulk temperature rise	6.93 K
Heat-up time constant τ	1.04 s

6 Conclusions

A three-dimensional transient finite-volume model of a showerhead injector operating on hydrogen peroxide was developed and exercised. For the baseline 316L configuration the simulation predicts:

1. a strongly non-uniform gas-side face temperature, from $\sim 393\,°C$ at the cooled channel wall to $\sim 849\,°C$ between orifices (a ~ 457 K in-plane gradient);
2. a fast thermal response of the critical gas-side face, with a heat-up time constant $\tau \approx 1.0$ s;
3. a manifold side that stays cold ($\sim 61\,°C$) thanks to the HTP cooling, while the deep through-thickness equilibrium is approached only slowly because of the low conductivity of stainless steel.

The framework is fully parametric, so material, mass flow, gas temperature and geometry can be varied directly in the source code. The largest modelling uncertainty is the gas-side film coefficient h_{gas}, to which the steady-state temperature is nearly proportional; substituting a Bartz-

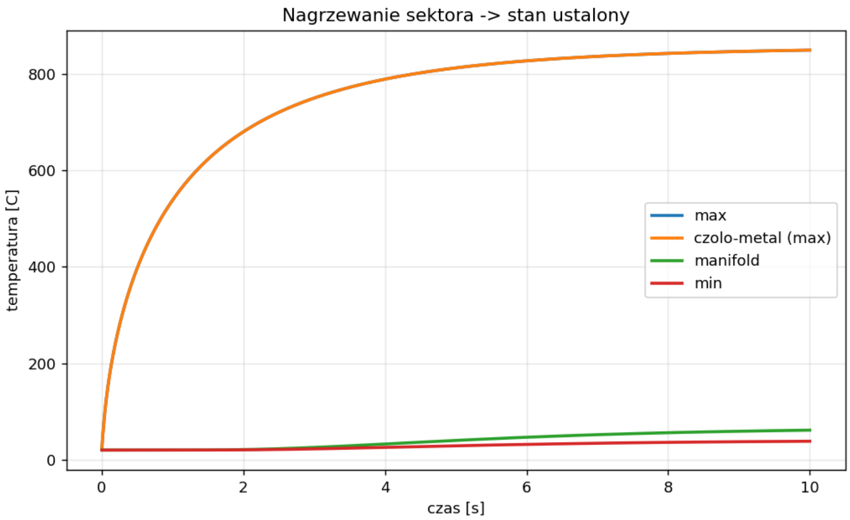


Figure 3: Transient temperature histories of the characteristic probes; the dashed line marks the whole-body steady-state criterion.

type correlation or a measured heat flux is the natural next step, together with temperature-dependent properties and a body-fitted (non-staircase) orifice mesh.

References

[1] D. K. Huzel and D. H. Huang. *Modern Engineering for Design of Liquid-Propellant Rocket Engines*. AIAA, 1992.

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[3] M. Ventura and E. Wernimont. *History of the Hydrogen Peroxide Bipropellant*. AIAA Joint Propulsion Conference, 2007.

[4] G. P. Sutton and O. Biblarz. *Rocket Propulsion Elements*. 9th ed., Wiley, 2016.